

Summary

10.10

Added a module called "tap (transport as plugin)" in Signal, changing the transmission of Signal ciphertext from Signal Server to a user-configured provider. Used in the following cases:

1. Private chat:

A initiates "use tap mode", generates a read-only token for A's configured cloud_A, and sends it to B.

After B accepts the request, B also sends a read-only token (of cloud_B) to A.

Then, tap mode is activated:

A sends messages by uploading ciphertext to cloud_A.

B periodically pulls ciphertext from cloud_A and decrypts it.

Similarly, B can send messages via Cloud B, and A pulls from Cloud B using token_B

2. Group chat:

A initiates "use tap mode", generates a read-only token for A's configured cloud_A, and sends it to all members. The group enters "Proposing" state.

After receiving the request, B agrees and sends a read-only token for cloud_B to all members.

...

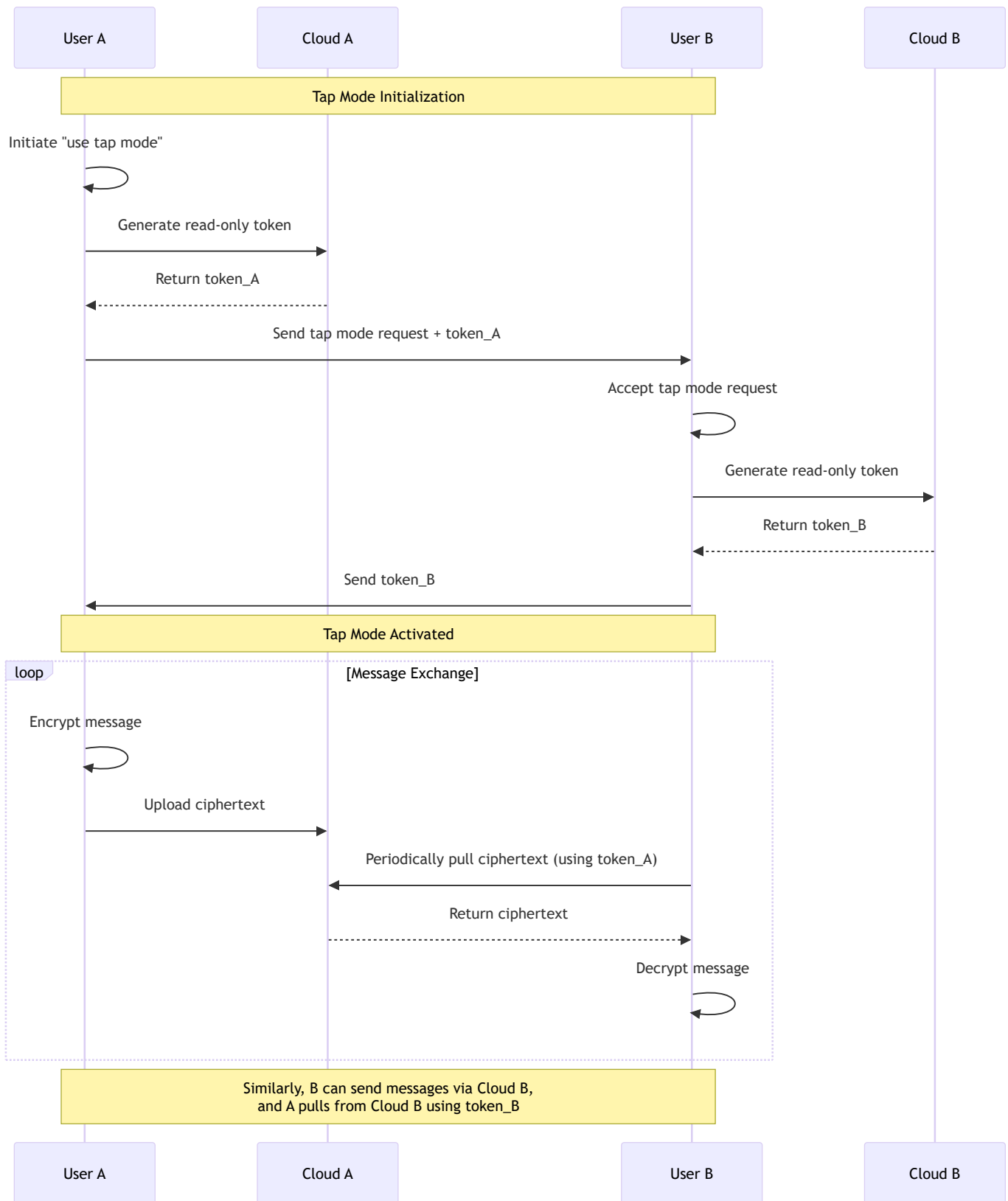
Once all members' tokens are locally available, the group enters tap mode.

A sends messages by uploading ciphertext to cloud_A.

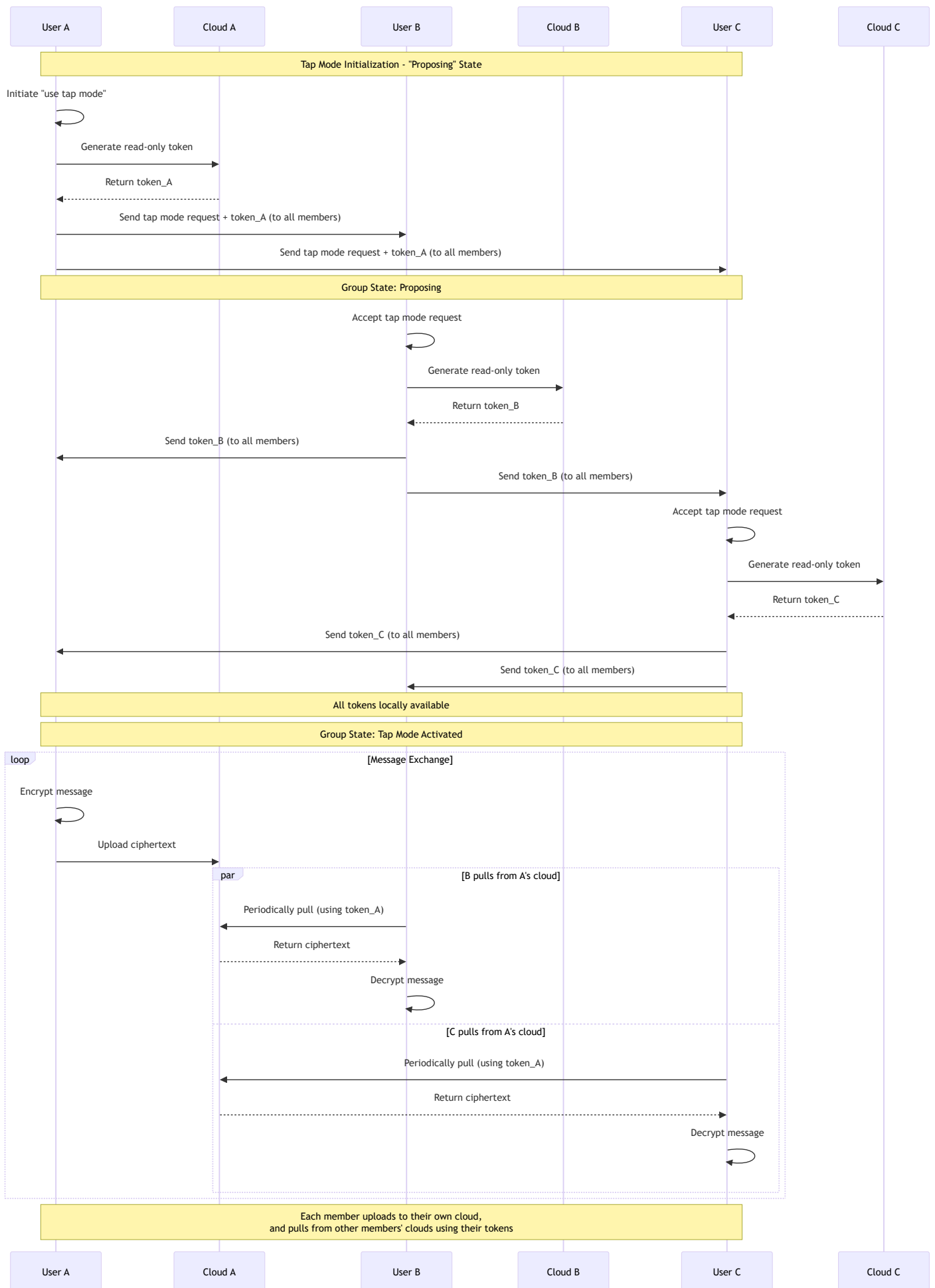
A periodically pulls ciphertext from other members' clouds, then decrypts them.

Diagrams

1. Private chat:



2. Group chat:



Some notes

1. An abstract class has been added; the program no longer directly depends on Tencent or AWS services.
Additional storage providers can be added in the future.
2. Only the transport layer was changed; the encryption feature remains unchanged.
3. One finding: Signal's two-person chat still uses Double Ratchet encryption, not Sender Key as used in other group chats.